

EduGenie AI

02 – Requirement Analysis

1. Introduction

Requirement analysis identifies what the EduGenie system should do and what resources are needed to develop and run the application. The system is designed as an AI-powered educational assistant for students.

2. Problem Statement

Students may have difficulty understanding complex topics, summarizing large amounts of study material, creating practice questions, and following a structured learning plan. EduGenie is proposed to provide these learning activities through a simple web application.

3. Functional Requirements

The functional requirements describe the main operations that EduGenie must provide.

- Ask Questions – Students can enter educational questions and receive clear answers.
- Explain Concepts – The system can explain a given concept in simple, beginner-friendly language.
- Summarize Text – The system can summarize supplied educational content and identify key points.
- Generate Quiz – The system can create multiple-choice questions with four options, answers, and explanations.
- Generate Learning Path – The system can create a structured learning plan based on topic, level, and duration.

4. Non-Functional Requirements

- Usability – The interface should be simple and easy for students to use.
- Performance – Requests should be processed within a reasonable time.
- Reliability – The application should handle errors and unavailable AI responses appropriately.
- Security – API keys and other sensitive configuration values should not be exposed in the source code.
- Maintainability – The application should use a clear structure so that features can be updated easily.

5. Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM recommended
- Keyboard and mouse
- Internet connection

6. Software Requirements

- Windows, macOS, or Linux operating system
- Python 3.x
- FastAPI
- Google Gemini API
- HTML, CSS and JavaScript
- Web browser
- Virtual environment for Python dependencies

7. User Requirements

- The user should be able to access the application through a web browser.
- The user should be able to enter questions or educational text.
- The user should be able to select or provide information required for quizzes and learning paths.
- The user should be able to understand the generated educational responses.

8. System Requirements

The system requires a running FastAPI application, the required Python packages, a valid Gemini API configuration for AI-powered features, and an internet connection for communication with the AI service.

9. Expected Output

- Clear educational answers
- Simple concept explanations
- Useful text summaries
- Structured quizzes
- Personalized learning paths

10. Conclusion

The requirement analysis defines the functional, non-functional, hardware, software, and user requirements of EduGenie. These requirements provide a clear foundation for the design and development phases.